Drastic prime conductor gap, round 3: paired targets

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Measured on public synthetic prime-field curves, 2026-10-12 UTC. Classification: boundary control and counted-work accounting.

Round 3 adds the paired target law `planted_shared`: the same scalar `k` on both curves of a pair (their `r` is equal), so the per-index quotient removes the between-target variance of round 2's independent draws. The D = -3 pairs re-measure under it with 64 workloads; a fresh D = -4 (j = 1728, cofactor 2) pair measures the sqrt(2) rung with the new `rho.quartic`; a D = -19 pair at ell = 4067213 (32x the 131101 the programme started from) sweeps the gap under the theorem.

family	ell	prediction	same-method paired (99%)	best-method paired (99%)
d3	1016839	sqrt(3) = 1.7321	**1.08144** (0.782885 to 1.47188)	**1.85905** (1.36982 to 2.50032)
d3	2033609	sqrt(3) = 1.7321	**0.931540** (0.696319 to 1.24097)	**1.91641** (1.39821 to 2.61342)
d4	1016839	sqrt(2) = 1.4142	**0.942538** (0.658469 to 1.40884)	**1.13648** (0.745565 to 1.95959)
d19	4067213	1 (the generic theorem at a 30x gap)	**1.01555** (0.646067 to 1.63847)	not run

The D = -3 sqrt(3) confirms under the paired law: both best-method 99% intervals exclude one. The same-method controls sit at one everywhere (the pairing shows it directly: the crater and floor solve the same scalars for the same counted cost). The D = -19 cell holds the generic theorem at a 32x gap.

The sqrt(2) rung under-realizes

The D = -4 pair's within-crater quotient reads 1.2058 [0.715 to 2.037] against the predicted sqrt(2) = 1.4142, and its best-method quotient 1.1365 [0.746 to 1.960] does not exclude one at 32 workloads: the quartic walk realizes about 1.2 of its fold where the cubic realizes its full sqrt(3). The walk's look-ahead is tuned for the negation map's 2-cycles; a four-class walk makes longer fruitless cycles likelier, and the counted rotations add one multiplication per canonicalisation. Walk tuning for A = 4 is the open engineering item; the mechanism is not in doubt (the D = -3 cubic and the rho-neg controls bracket it).

Recorded execution and replay

- **d4-ell1016839-rho**, audit OK: true; verified records: 256; exact measured replays: 128. [audit](sessions/d4-ell1016839-rho.audit.json), [sealed session](sessions/d4-ell1016839-rho/session.json).
- **d4-ell1016839-auto**, audit OK: true; verified records: 64; exact measured replays: 32. [audit](sessions/d4-ell1016839-auto.audit.json), [sealed session](sessions/d4-ell1016839-auto/session.json).
- **d3-ell1016839-rho**, audit OK: true; verified records: 512; exact measured replays: 256. [audit](sessions/d3-ell1016839-rho.audit.json), [sealed session](sessions/d3-ell1016839-rho/session.json).
- **d3-ell1016839-cubic**, audit OK: true; verified records: 128; exact measured replays: 64. [audit](sessions/d3-ell1016839-cubic.audit.json), [sealed session](sessions/d3-ell1016839-cubic/session.json).
- **d3-ell2033609-rho**, audit OK: true; verified records: 512; exact measured replays: 256. [audit](sessions/d3-ell2033609-rho.audit.json), [sealed session](sessions/d3-ell2033609-rho/session.json).
- **d3-ell2033609-cubic**, audit OK: true; verified records: 128; exact measured replays: 64. [audit](sessions/d3-ell2033609-cubic.audit.json), [sealed session](sessions/d3-ell2033609-cubic/session.json).
- **d19-ell4067213-rho**, audit OK: true; verified records: 256; exact measured replays: 128. [audit](sessions/d19-ell4067213-rho.audit.json), [sealed session](sessions/d19-ell4067213-rho/session.json).

Reproduce

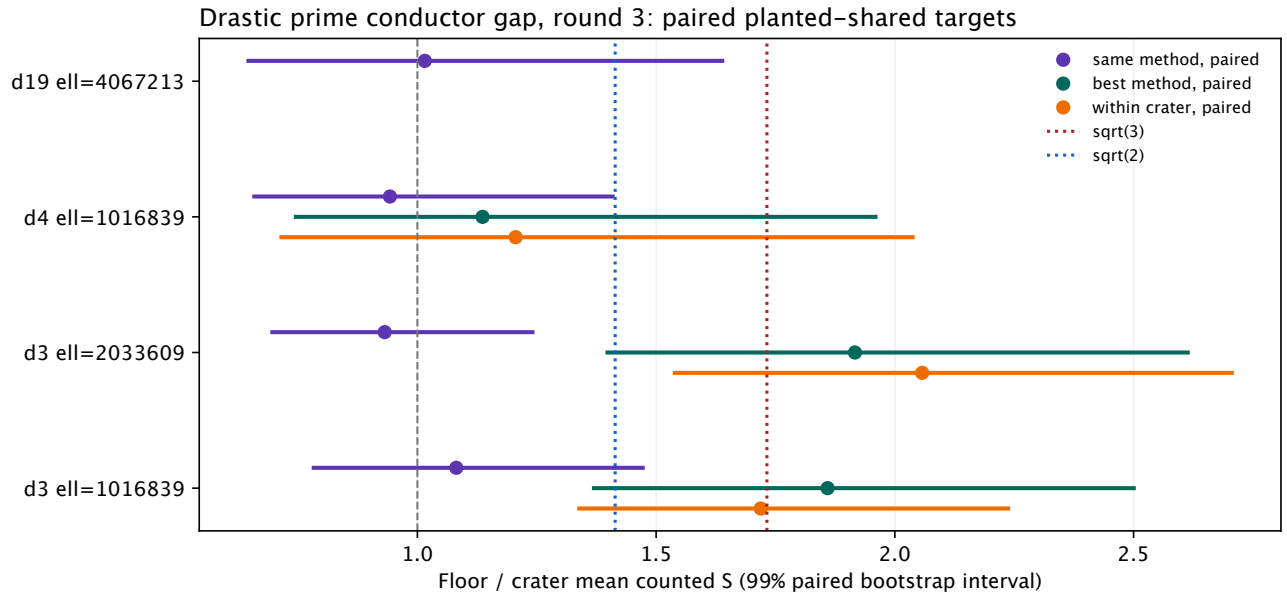
`drastic_gap_round3_pairs --out pairs/pairs.json`; `drastic_gap_round3_specs --pairs pairs/pairs.json --round2 ../drastic_prime_conductor_gap_round2_20261012/pairs/pairs.json --out specs`; `run_sessions.sh`; analysis: `drastic_gap_round3_analyze --sessions research/drastic_prime_conductor_gap_round3_20261012/sessions --out research/drastic_prime_conductor_gap_round3_20261012/analysis`. Frozen seed 20261012; 20,000 resamples.

Requirement status

requirement	status	evidence or remaining gap
Higher-confidence intervals	measured	paired planted_shared law, 64 workloads; round-2 quotients re-measured
The sqrt(2) rung	measured, under-realized	rho.quartic realizes ~1.2 of sqrt(2); walk tuning open
The 32x gap sweep	measured	same-method = 1 at ell = 4067213
Tier-1 Linux wall times	blocked	tools/isolated_bench.py needs a Linux host with PSI and taskset; none in this environment
External-host replay	blocked	same host class
Root-finding PDP across the gap	deferred	#1729's harness is not yet an ecbench arm
The cubic fold of the MITM table	frozen as round 4	PROTOCOL.md E4

Evidence and visual scope

[Protocol](PROTOCOL.md), [native tables](analysis.md), [machine-readable analysis](analysis.json), [pair search output](pairs/pairs.json), [report PDF](report.pdf). The scoreboard receives a bounded-count panel; no frontier or leaderboard promotion. The Conductor service was unreachable at session time; the check was attempted before editing.



The same planted scalar on both curves; joint bootstrap over indices, 20,000 resamples. Count diagnostics; no wall or hardness claim.